

Program: Diploma in Microelectronics	
Course Code: 6497	Course Title: FPGA Lab
Semester: 6	Credits: 2.5
Course Category: Program Core	
Periods per week: 4 (L:0, T:1, P:3)	Periods per semester: 60

Course Objectives:

- To Implement and test the functional aspects of digital electronic circuits FPGA (Xilinx. Altera)

Course Prerequisites:

Topic	Course code	Course name	Semester
Combinational & Sequential circuits		Digital Electronics	3
CMOS		VLSI Design	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Implement various Combinational Circuits in different levels of Abstraction	19	Applying
CO2	Implement various Sequential Circuits in different levels of Abstraction	19	Applying
CO3	Implement state machine (mealy and Moore) on FPGA	8	Applying
CO4	Implement basic gates using Switch level modelling	8	Applying
	Lab Exam	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3	3	3		3	2
CO2	3	3	3	3		3	2
CO3	3	3	3	3		3	2
CO4	3	3	3	3		3	2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Implement various Combinational Circuits in different levels of Abstraction		
M1.01	Implement Adders and Subtractors on FPGA	4	Applying
M1.02	Design a 2 bit Comparator and implement it on FPGA	3	Applying
M1.03	Design a 4 bit Code converter and implement it on FPGA	4	Applying
M1.04	Design a Multiplexer/Demultiplexer & decoder/Encoder logic circuit and implement it on FPGA	4	Applying
M1.05	Design a 4 bit parallel Adder/Subtractor and implement it on FPGA	4	Applying
CO2	Implement various Sequential Circuits in different levels of Abstraction		
M2.01	Implement the following flipflops on FPGA 1. SR Flip Flop 2. JK Flip Flop 3. D Flip Flop 4. T Flip Flop 5. Master Slave J K Flip Flop	7	Applying
M2.02	Design the following Synchronous counters and implement on FPGA 1. 3 bit up/down counter 2. Mod-10 up Counter 3. Mod-13 down counter	5	Applying

M2.03	Design the following Aynchronous counters and implement on FPGA 1. 4 bit up/down counter 2. Mod-7 up Counter 3. Mod-11 down counter	3	Applying
M2.04	Implement SISO and PISO shift register on FPGA	4	Applying
	Lab Exam	3	
CO3	Implement state machine (mealy and Moore) on FPGA		
M3.01	Develop a Mealy model to generate a given sequence	4	Applying
M3.02	Develop a Moore model to generate a given sequence	4	Applying
CO4	Implement basic gates using Switch level modelling		
M4.01	Develop a CMOS Inverter	2	Applying
M4.02	Implement CMOS NAND/AND	2	Applying
M4.03	Implement CMOS NOR/OR	2	Applying
M4.04	Implement CMOS XOR/XNOR	2	Applying
	Lab Exam	3	

****Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can-do open-ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation and a score of 15 marks. Students should prepare a separate report on an open ended experiment of their choice.)

Example:

1. Implement Switch level modelling of Half Adder on FPGA
2. Implement Switch level modelling of Full Adder on FPGA
3. Implement Switch level modelling of Half Subtractor on FPGA

Text /Reference:

T/R	Book Title/Author
T1	Digital System Design with FPGA: Implementation Using Verilog and VHDL Cem Unsalan, Bora Tar, MC Graw hill, 2017
T2	A Practical Guide for Simulation and FPGA Implementation of Digital Design, Bekkay Hajji , Adel Mellit , Loubna Bouselham, Springer,2024
R1	Brown S. and Vranesic Z., Fundamentals of Digital Logic with VHDL Design, 2/e, TMH.2008
R2	Perry D. L., VHDL Programming by Example, 4/e, TMH, 2008.
R3	Roth C.H., Digital System Design Using VHDL, Cengage Learning, 2008
R4	T.R.Padmanabhan,B BalaTripura Sundari, Design Through Verilog HDL, Wiley2009.
R5	FPGA prototyping by Verilog examples–PONGP Chu-Willey

Online Resources:

Sl. No	Website Link
1	https://vhdlwhiz.com/terminology/combinational-logic/
2	https://www.xilinx.com/support/documentation-navigation/self-paced-tutorials/see-all-tutorials.html
3	https://www.eecs.umich.edu/courses/doing_dsp/handout/vhdl-tutorial.pdf
4	https://personal.utdallas.edu/~mtacca/courses/EE3120/Spring-2007/Xilinx_tutorial.pdf